3 A hollow cylinder, which is open at both ends, has a radius of $(3.0 \pm 0.1)\,\text{cm}$ and a length of $(15.0 \pm 0.1)\,\text{cm}$.

What is the value, with its absolute uncertainty, of the surface area of the cylinder?

A $(280 \pm 10)\,\text{cm}^2$

B $(282.7 \pm 0.2)\,\text{cm}^2$

C $(420 \pm 30)\,\text{cm}^2$

D $(424.1 \pm 0.3)\,\text{cm}^2$